Individual Project Canvas (A3 page)

S226 PRT681/PRT585 SOFTWARE ENGINEERING: PRACTICE

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1. PROJECT TITLE

AI-Powered Online CV Maker

2. VISION

To develop an intelligent web application that supports job seekers in generating professional, ATS-friendly CVs tailored to specific job descriptions using artificial intelligence. This platform aims to streamline the resume creation process by improving content relevance, formatting quality, and compatibility with applicant tracking systems.

3. PROBLEM STATEMENT

Many job applicants submit generic CVs that do not adequately satisfy employer expectations or Applicant Tracking System (ATS) requirements. Moreover, tailoring a CV for each job application is often time-consuming and difficult, creating a significant barrier to effective job search efforts.

4. OBJECTIVES

- Enable users to create and manage their professional profiles.
- Utilize artificial intelligence to analyze job descriptions.
- Automatically generate personalized CVs.
- Offer multiple professional CV templates.
- Allow users to export CVs in PDF format.

5. TECHNICAL KNOWLEDGE

- **Frontend:** React.js / Next.js
- **Backend:** .NET Web API
- **Database:** PostgreSQL
- **AI Integration:** Large Language Model (LLM) API for analyzing job descriptions and generating optimized CV content
- **Authentication:** JWT Authentication
- **Version Control:** Git & GitHub
- **Deployment:** Docker / Azure / Render (or similar cloud platform)

6. MARKET KNOWLEDGE

The recruitment market increasingly depends on Applicant Tracking System (ATS) software to screen and filter job applications. As a result, job seekers must prepare customized CVs that align closely with job descriptions to improve their chances of securing interviews. In response to this demand, AI-powered resume builders are experiencing rapid growth as they offer a faster and more effective approach to job application preparation.

7. INNOVATION

- AI analyzes job descriptions automatically.
- Generates ATS-friendly CVs.
- Suggests missing skills and keywords.
- Recommends professional wording for work experience.
- Supports multiple modern CV templates.
- Reduces CV preparation time significantly.

8. INTERNAL STAKEHOLDERS

- Project Developer
- Academic Supervisor
- University
- System Administrator
- Development Team (future)

9. EXTERNAL STAKEHOLDERS

- Job Seekers
- Employers
- Recruitment Agencies
- HR Professionals
- AI Service Provider
- Cloud Hosting Provider

10. ORGANIZATION

The project follows a layered architecture:

- Presentation Layer (React / Next.js)
- Business Logic Layer (.NET API)
- Database Layer (PostgreSQL)
- AI Service Layer (LLM API)

11. PRIORITIZATION

High Priority

- User Registration & Login
- User Profile Management
- AI CV Generation
- PDF Download

Medium Priority

- Multiple CV Templates
- CV Editing
- Dashboard

Low Priority

- Cover Letter Generator
- Interview Preparation
- Portfolio Builder

12. PRODUCT FEEDBACK

Feedback will be collected from:

- Students
- Career advisors
- Recruiters
- Supervisors

Evaluation criteria:

- Ease of use
- CV quality
- AI accuracy
- User satisfaction
- ATS compatibility

13. PROCESS FEEDBACK

Development will follow an Agile approach:

- Requirement Analysis
- UI/UX Design
- Backend Development
- Frontend Development
- AI Integration
- Testing
- Deployment
- User Feedback
- Continuous Improvement

14. READY (DEFINITION OF DONE)

The project is considered complete when:

- Users can register and log in.
- User profiles are stored securely.
- AI successfully analyzes job descriptions.
- Customized CVs are generated.
- Users can edit and download CVs as PDF.

15. TIMELINE

Week	Activity
1-2	Requirement Analysis & Design
3-4	Backend Development (.NET)
5-6	Frontend Development (React/Next.js)
7	AI Integration
8	Testing & Bug Fixes
9	Documentation & Deployment

16. DELIVERABLES

- Responsive Web Application
- PostgreSQL Database
- REST API (.NET)
- AI CV Generator
- Documentation
- Source Code
- Deployment

17. SUCCESS CRITERIA

- Generate professional ATS-friendly CVs in under 2 minutes.
- Improve CV relevance to job descriptions.
- Positive user feedback.
- Secure and responsive application.
- Easy-to-use interface.